MIL-B-36492
13 June 1966

MILITARY SPECIFICATION

BED, ORTHOPEDIC

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a turning frame for use in orthopedic, neurosurgical, and gynecological services where immobilization of the spine, with or without hyperextension is required, and where frequent turning of the patient is necessary for proper nursing care. The turning frame shall be suitable for use in aeromedical evacuation.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-200/8	Aluminum Alloy Bar, Rod, Shapes and Tube, Extruded 6061 and 6062.
CCC-C-419	Cloth, Duck, Cotton, Unbleached, Plied Yarns, Army and Numbered.
CCC-C-430	Cloth, Cotton, Sheeting.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6530

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STANDARDS

Federal

FED-STD-66	Steel, Chemical Composition and Hardenability.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), prior to the start of the production of the contract quantity, one preproduction sample of the turning frame shall be assembled, completely finished, and tested by the contractor. The sample shall be submitted, together with the contractor's inspection and test reports, to the contracting officer for examination and testing to determine compliance with the requirements of this specification. Written approval of the materials, construction, and performance of the sample shall be obtained from the contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Further production of the item by the contractor, prior to this approval, shall be at the contractor's risk. Accepted preproduction sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the production sample is not approved, the contractor shall submit a new sample for approval, or modify the existent sample so as to remove the deficiencies cited.

3.2 Materials. All materials used in fabrication of the turning frame shall be as specified herein. Commercial tolerances on gage sizes shall apply (see Table II).

3.2.1 Aluminum Alloy. Aluminum tubing shall be temper T6, conforming to QQ-A-200/8 for grade 6062.

3.2.2 Steel. Seamless steel tubing shall conform to FED-STD-66 for MT1010 carbon steel.

3.2.3 Cotton sheeting. Cotton sheeting shall conform to CCC-C-430 for type II, class 2, style A.

3.2.4 Canvas. Canvas shall be unbleached cotton, duck, conforming to CCC-C-419, for type I, duck number 11.

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3.3 Construction. The turning frame shall consist of a supporting frame, an anterior and a posterior frame, a supporting, turning and locking mechanism, two armboards, and a tray holder and reading board. Also included shall be one inch thick, firm polyether foam mattresses, sheets, canvases, two head support bands, an adjustable facepiece with three extra sets of forehead and chin supports, and a detachable foot support assembly. In the setup position, the turning frame shall be 87 inches long and 22 inches wide. The foot end shall be open to accommodate patients of any size. The parts used in fabrication of the turning frame shall be as listed in Table I and corresponding Figure 1. Parts made of tubing shall be of the gages and sizes listed in Table II.

3.3.1 Supporting frame. The supporting frame shall consist of a round steel tubular wraparound base runner and square steel tubular head end and foot end supports. Means shall be provided whereby the supporting base runner can be securely attached to the orthopedic bed cart (see 6.1). The foot end support shall be attached to the circular supporting, turning and locking mechanism.

3.3.2 Supporting, turning and locking mechanism. The supporting, turning and locking mechanism shall consist primarily of two radial frames of steel tubing. This mechanism shall provide support between the square tubular foot end support and the anterior and posterior frames, and shall provide adjustment between the frames to suit the thickness of the patient.

3.3.3 Anterior and posterior frames. The anterior and posterior frames shall consist of aluminum tubing. The two frames shall be attached to the head and foot end supports in such a manner that one person can rotate the frames 180° to reverse a patient's position. When turning has been accomplished, a locking pin in the turning mechanism shall secure the anterior and posterior frames. No permanent set shall develop, and the force required to rotate the frames shall not exceed 16 pounds lineal pressure when the frames are tested as specified in 4.4.1

3.3.4 Armboards. The armboards shall be made of 10 x 20 x 1/4 inch plywood. The armboards shall be capable of being adjusted up, down, forward, back, tilted, and locked by the patient, at any one of four preset positions, in the horizontal plane.

3.3.5 Tray holder and reading board. The tray holder and reading board shall be one piece of 0.048 inch (No. 18, United States Standard gage) thick carbon steel finished in gray hammertone enamel. Overall dimensions shall be 22-3/4 x 21-1/8 x 3-1/2 inches. It shall fit between the long sides of the supporting frame base runner when the frame is mounted on the orthopedic bed cart.

3.4 Weight. Net weight of the assembled unit less soft goods shall not exceed 65 pounds.

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TABLE I. List of Parts

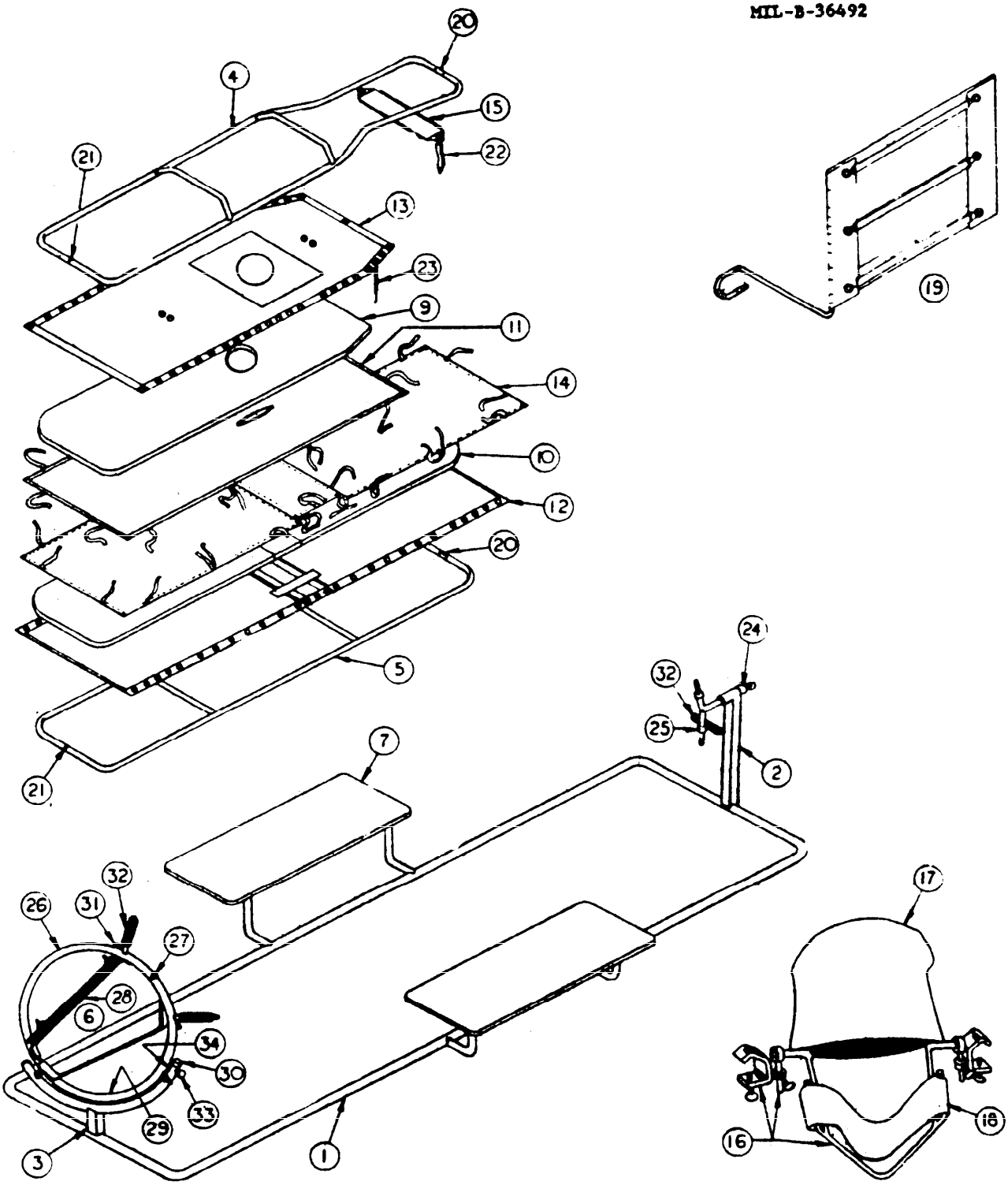
Part No.	Description	Req'd
1	Base Runner, Supporting Frame	1
2	Head End Support	1
3	Foot End Support	1
4	Anterior Frame	1
5	Posterior Frame	1
6	Supporting, Turning, and Locking Mechanism	*
7	Arm Board	2
8	Tray Holder and Reading Board (not shown)	1
9	Anterior Mattress	1
10	Posterior Mattress	1
11	Anterior Sheet	1
12	Posterior Sheet Set	1
13	Anterior Canvas	1
14	Posterior Canvas Set	1
15	Head Support Band	2
16	Adjustable Facepiece	1
17	Forehead Support	4
18	Chin Support	4
19	Foot Support Assembly, Adjustable	1
20	Swivel, Head	2
21	Swivel, Foot	2
22	Evertaut Assembly (short)	4
23	Evertaut Spring (short)	47
24	Traction Collar Assembly	1
25	Swivel	1
26	Anterior Radial Frame	1
27	Circle Lock	1
28	Anterior Support	1
29	Posterior Radial Frame	1
30	Plastic Stop Plug	2
31	Stop Button	2
32	Hand Grip	3
33	Turning Lock	1
34	Lock Pin	1

* One assembly consisting of part numbers 26 through 33.

TABLE II. Tubing - Gage, Size, and Wall

Part Name	Birmingham or Stubs' Gage	Size (inches)	Wall Thickness
Foot End Support	11	1-1/4 square	.120
Head End Support	12	1-1/4 square	.109
Base Runner	14	1 diameter	.083
Radial Frames	14	1 diameter	.083
Anterior Frame	14	1-1/4 diameter	.083
Posterior Frame	14	1-1/4 diameter	.083

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WEDGE TURNING FRAME

FIGURE 1

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3.5 Finish.

3.5.1 Aluminum tubing. All aluminum tubing shall be given an anodized finish in conformance with MIL-A-8625.

3.5.2 Steel tubing. All carbon steel tubing shall be chromium plated over nickel over copper.

3.5.3 Hardware. All screws, bolts, nuts, lock pins, and all other steel parts except steel tubing shall be heavily nickel plated over copper or shall be cadmium plated.

3.5.4 Plywood. All plywood parts shall be sanded smooth, sealed with a wood sealer, and then sprayed with a clear lacquer.

3.6 Identification marking. Each turning frame shall have a metal nameplate securely attached and permanently marked with the following:

Manufacturer's name or registered trademark
Serial number
Contract or purchase order number
Federal stock number
Letters U.S.

Marking secured by adhesives shall not be used.

3.7 Service data. The contractor shall furnish, with each unit, two copies of service data containing complete instructions for installation, operation, maintenance, and a list of component parts and accessories. Included shall be a traction chart for determining the amount of cart elevation necessary for the pounds of traction desired. The service data shall include such detailed information which shall permit ready installation and maintenance of the item. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

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3.8 Workmanship. Workmanship shall be first class throughout. The turning frame shall be free from defects which detract from its appearance or may impair its serviceability.

3.8.1 Welding. All required joints of the frames shall be continuously welded or joined by brazing. Corners shall be rounded and ground smooth to blend with adjacent areas. Welds shall be durable and sound in every detail. Prior to welding, all components shall be properly aligned in their proper positions to insure a straight, rigid, and level assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Total (major and minor combined) defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for the operational test shall be in accordance with inspection level S-4 of MIL-STD-150. The acceptance number shall be zero regardless of lot size taken.

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4.3 Examination. The frames shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Catgories and defects *

Major
101 Turning frame design not as specified.
102 Moving parts do not move freely.
103 Threads not free of stripping.
104 Means not provided to attach support runner to the cart.
105 Nameplate not supplied, incomplete, or illegible.
106 Metal parts not free of cracks, tears, ruptures, or deformed surfaces.
107 Welds not free of cracks; welds broken or not sound.
108 Service data missing or does not include all specified information.
109 Exposed or accessible edges, corners, or surfaces not free of burrs, sharpness, and sharp projections.

Minor
201 Plated finish not free of blisters, pits, peeling, or visible porosity.
202 Commercial finish (No. 4) not uniform, not free of scratches, rough grind marks, and other defects.
203 Finish not as specified.
204 Welds not continuous, not clean, not free of pitting, spatter, and discoloration.
205 Exposed welds not ground smooth to blend with adjacent surfaces.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The frames shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Operational test. With the posterior frame in its normal horizontal position, sand bags totaling 250 pounds in weight shall be equally distributed and secured to the frame. After placing the anterior frame over the sand bags, the assembled orthopedic turning frame shall be repeatedly rotated to all possible positions to determine compliance with 3.3.3.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One bed, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit preparation. Each unit shall be disassembled as far as practicable. The anterior and posterior mattress shall be wrapped with kraft paper and wrapping secured by tape. The anterior and posterior sheet and canvass shall be folded to an appropriate size wrapped with kraft paper and wrapping secured by means of tape. The canvass, sheets, and mattress shall be further packaged in a carton of appropriate size conforming to PPP-B-636. The anterior, posterior, and base frames shall be wrapped with kraft paper to prevent marring or scratching and wrapping secured by tape. Arm board and miscellaneous items shall be wrapped with kraft paper and packaged in cartons of appropriate size conforming to PPP-B-676, or PPP-B-636, except that commercial colors shall be acceptable. Cartons shall be designed with interior fittings or contain sufficient cushioning material to insure against damage of or to the contents. Closure of cartons shall be adequate to prevent spilling of contents under normal handling.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Each unit prepared as specified in 5.1.2.1 shall be blocked, braced and cushioned in an exterior container designed for a type 3 load constructed in accordance with PPP-B-601, table II, using type III, class 1 plywood. Blocking, bracing and cushioning shall be adequate to prevent shifting of or damage to the contents.

5.2.1.2 Case liner. Each level A box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 Closure. Closure of boxes shall be in accordance with appendix of applicable box specification.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The orthopedic turning frame is intended for mounting on CART, ORTHOPEDIC BED, Stryker.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).
(c) When preproduction sample is not required (see 3.1).
(d) Where preproduction sample shall be sent (see 3.1).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DN

Review activites:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

CATEGORY N

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-B-36492

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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PHILADELPHIA, PA. 19101

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